

Jiarun Liu

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SUMMARY

I'm a 3DCV researcher and algorithm engineer, mainly focusing on world models and world action models in autonomous driving scenarios.

Before that, I got my M.S. degree in Zhejiang University, advised by [Prof. Guofeng Zhang](#), studying on SLAM, visual localization and neural reconstruction.

I am interested in 3D computer vision and generative models, as well as world action models and applications in autonomous driving and embodied AI.

EDUCATION

2022 - 2025	M.S. in Software Engineering at Zhejiang University	(GPA: Top 5%)
2018 - 2022	B.S. in Digital Media at Zhejiang University	(GPA: Top 5%)

SELECTIVE AWARDS AND HONOR

Outstanding Graduates at ZJU, 2022 & 2025

Tianzhou Chen Scholarship of ZJU, 2023

Yongping Scholarship of ZJU, 2020

INTERNSHIP & WORK EXPERIENCE

Senior AI Algorithm Engineer, NIO Inc. Jan 2026 - Now

- Research on foundation model. My main research direction is multi-modal generation and WAM.
- Research on visual-native 4D world model, mainly focus on geometric representation autoencoder (G-RAE) for high-dimensional representation and generation.
- Work on end-to-end planner pretraining, mainly focus on auxiliary loss design and joint-prediction MoT architecture design.
- Research on multi-modal world action model.

Algorithm Engineer, Cainiao Inc., Alibaba Group Mar 2025 - Jan 2026

- Work on world model and diffusion model. Currently, we've investigated in many directions, including DiT-based multi-view AD video generation, SVD-based synthetic video harmonization and more.
- Work on Gaussian Splatting-based sensor simulation system. I mainly focused on street scene reconstruction, editing (insertion and inpainting) and scene expansion with diffusion guidance.
- Research on multi-modal learning, mainly on text-image-3DGS feature alignment.
- Research on 3D reconstruction, mainly on visual reconstruction and depth estimation.

Research Intern, Cainiao Inc., Alibaba Group Mar 2024 - Mar 2025

- Work on SLAM & point cloud registration.
- Research on Gaussian Splatting and scene inpainting.

Research Intern, SenseTime Group Mar 2023 - May 2024

- Create a large-scale indoor localization benchmark, along with an open-source dataset.
- Research on mesh-based landmark localization.
- Maintainer of [XRLocalization](#), an open-source visual re-localization framework.

PROJECTS

E2E Planner with Auxiliary Prediction Supervision

Work done in NIO

I am one of the core contributors of NIO's End-to-end planner algorithm. My work focus on joint visual-action prediction, the design of Mixture-of-Transformer (MoT) structure, the future prediction auxiliary loss and the integration to pretraining pipeline.

N-WFM: Vision-Native 4D World Foundation Model

Work done in NIO

I am the core contributor of NIO's World Foundation Model. My work focus on 4D generation based on geometry-grounded latent space¹. The work shows great expansion on latent diffusion model in geometry space.

Physical Time Understanding in Video Generation

Work done in NIO

I guided interns to carry out work on physical time understanding and control in video diffusion model. 2 Papers under review [1, 2].

Multi-modal Joint Diffusion for Reconstruction and Detection

Work done in NIO

I guided interns to carry out work on multimodal joint representation and diffusion, using a unified representation space and a generative framework to simultaneously perform high-quality dynamic scene reconstruction and 3D detection. 1 Paper under review [3].

GS-based Sensor Simulation System

Work done in Cainiao

We build an industrial-grade multi-sensor simulation system for unmanned vehicle, which provides more than 210k+ frames training data for perception models. I mainly focus on Neural GS framework building, multi-camera optimization, object generation, scene editing, and harmonization. 5 Papers have been accepted [4, 5, 6, 7].

3DGS Database & Multi-modal Retrieval

Work done in Cainiao

I built the AIGC 3D assets pipeline for our simulation system, including 200+ objects and 100+ scenes. I also worked on multi-modal alignment for text-image-3D feature alignment and retrieval. 1 Paper has been accepted [8].

Visual-based 3D Reconstruction & MVS

Work done in Cainiao

I joined the visual reconstruction team for 2 months and worked in spatial-temporal consistency MVS pipeline for better depth estimation. I also built the depth GT refinement pipeline for data curation. 1 Paper has been accepted [9].

Visual Localization for AR Application

Work done in SenseTime

I worked on mesh-based visual localization for outdoor landmark AR in anywhere. I also built a large-scale indoor localization benchmark. 1 Paper has been accepted [10].

PUBLICATIONS²

- [1] Hongbo Liu, Chunyong Hu, Hui Lu, **Jiarun Liu**, Jindong Tian, Lipei Zhang, Kai Zou, Dian Zheng, Shengjie Zhao, Sheng Yang, and Shaoqing Ren. "ScenePilot: A Streaming Video World Model with Camera-Time Control". In: *AAAI Conference on Artificial Intelligence*. 2027.

¹This work was done 3 months before Tencent ARC's [GAE](#) was published, but kept unreleased for some internal reason.

²* denotes equal contribution, † denotes corresponding author.

- [2] Jindong Tian, Lipei Zhang, Chunyong Hu, **Jiarun Liu**, Hongbo Liu, Hui Lu, Chenjuan Guo, Bin Yang, Sheng Yang, and Shaoqing Ren. “ Δt Forcing : VideoPredictionConditionedonPhysicalTime”. In: *AAAI Conference on Artificial Intelligence*. 2027.
- [3] Li-Heng Chen, Haokai Pang, Chengye Su, **Jiarun Liu**, Qifeng Chen, Ziqian Ni, Jianxin Huang, Shi-Sheng Huang, Hongbo Fu†, and Sheng Yang†. “USR-Drive: Unified Generative Driving Scene Representation via Joint Denoising of 3D Gaussians and Boxes”. In: *AAAI Conference on Artificial Intelligence*. 2027.
- [4] Xianming Zeng*, Sicong Du*, Qifeng Chen*, Lizhe Liu, Haoyu Shu, Jiaxuan Gao, **Jiarun Liu**, Jiulong Xu, Jianyun Xu, Mingxia Chen, Yiru Zhao, Peng Chen, Yapeng Xue, Chunming Zhao, Sheng Yang†, and Qiang Li. “Industrial-Grade Sensor Simulation via Gaussian Splatting: A Modular Framework for Scalable Editing and Full-Stack Validation”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025.
- [5] Guo Chen*, **Jiarun Liu***, Sicong Du, Chenming Wu, Deqi Li, Shi-Sheng Huang†, Guofeng Zhang, and Sheng Yang†. “GS-RoadPatching: inpainting Gaussians via 3D Searching and Placing for Driving Scenes”. In: *ACM SIGGRAPH Asia*. 2025.
- [6] Sicong Du*, **Jiarun Liu***, Qifeng Chen, Hao-Xiang Chen, Tai-Jiang Mu, and Sheng Yang†. “RGE-GS: Reward-Guided Expansive Driving Scene Reconstruction via Diffusion Priors”. In: *IEEE International Conference on Computer Vision (ICCV)*. 2025.
- [7] Qifeng Chen, **Jiarun Liu**, Rengan Xie, Tao Tang, Sicong Du, Yiru Zhao, Yuchi Huo, and Sheng Yang†. “LiDAR-GS++: Improving LiDAR Gaussian Reconstruction via Diffusion Priors”. In: *AAAI Conference on Artificial Intelligence*. 2026.
- [8] **Jiarun Liu***, Qifeng Chen*, Yiru Zhao, Minghua Liu, Baorui Ma, and Sheng Yang†. “TIGaussian: Disentangle Gaussians for Spatial-Awared Text-Image-3D Alignment”. In: *International Conference on Learning Representations (ICLR)*. 2026.
- [9] Qihao Sun*, **Jiarun Liu***, Ziqian Ni*, Jianyun Xu, Tao Xie, Lijun Zhao, Ruifeng Li†, and Sheng Yang†. “LiDAR Prompted Spatio-Temporal Multi-View Stereo for Autonomous Driving”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.
- [10] Hailin Yu, **Jiarun Liu**, Zhichao Ye, Xinyu Chen, Ruohao Zhan, Yichun ShenTu, Zhongyun Lu, and Guofeng Zhang†. “Indoor large-scale panoramic visual localization dataset”. In: *Journal of Image and Graphics*. 2025.

SKILLS

English TOEFL 108 & CET-6 certified.

Cello Grade 10 certified. Experienced in band and orchestra.